

JOENSUU AP, Finland

WMO: 029290

Lat: 62.660N

Lon: 29.611E

Elev: 121

StdP: 99.88

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-24.1	-30.4	0.2	-27.4	-26.7	0.3	-24.1	8.0	-5.7	7.3	-5.3	1.3	300	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.9	18.4	24.9	17.3	23.0	16.5	19.7	24.3	18.6	23.0	17.5	21.6	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.2	21.6	16.9	12.3	20.7	15.9	11.4	19.6	56.9	24.1	53.2	22.9	49.9	21.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.6	6.1	-29.7	29.2	4.2	2.6	-32.7	31.1	-35.1	32.6	-37.5	34.1	-40.5	35.9		
(5)			-29.8	21.1	4.1	1.6	-32.8	22.2	-35.2	23.1	-37.6	24.0	-40.6	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-9.2	-9.0	-4.3	2.3	8.8	13.8	17.2	15.0	9.9	3.7	-1.7	-5.7		
	DBStd	10.35	7.63	7.26	4.90	3.86	4.73	3.81	3.37	3.19	3.37	3.95	4.75	6.72		
	HDD10.0	2971	594	533	443	232	79	8	1	2	43	198	351	487		
	HDD18.3	5466	852	766	701	481	296	143	61	111	253	455	601	745		
	CDD10.0	592	0	0	0	2	42	124	224	158	40	2	0	0		
	CDD18.3	44	0	0	0	0	1	8	27	8	0	0	0	0		
	CDH23.3	340	0	0	0	0	21	61	201	58	0	0	0	0		
	CDH26.7	55	0	0	0	0	1	8	38	7	0	0	0	0		
(14) Wind	WSAvg	3.0	3.0	2.9	3.0	2.8	3.1	3.0	2.8	2.7	2.9	3.1	3.3	3.3		
(15) Precipitation	PrecAvg	627	41	32	32	34	39	62	74	83	61	62	55	48		
	PrecMax	857	89	77	86	78	87	132	176	206	146	127	146	88		
	PrecMin	399	8	5	6	5	3	3	16	25	19	11	13	15		
	PrecStd	113	21	18	15	19	19	30	39	41	26	29	24	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.9	8.2	17.4	26.0	28.2	29.7	27.4	20.8	13.0	8.5	6.0		
		MCWB	2.3	2.4	4.0	10.6	14.6	17.3	20.3	19.0	15.7	11.2	7.8	5.5		
	2%	DB	1.8	1.9	5.1	13.5	22.4	25.2	27.6	24.9	18.2	11.3	6.9	4.0		
		MCWB	1.4	1.4	2.2	8.2	13.8	16.8	19.1	17.6	14.4	10.3	6.2	3.6		
	5%	DB	0.9	0.9	3.2	11.0	19.5	23.0	25.7	22.9	16.7	10.4	5.5	2.6		
		MCWB	0.7	0.5	1.1	6.3	11.9	15.8	18.3	16.8	13.7	9.4	5.1	2.1		
	10%	DB	0.0	0.1	1.9	8.7	17.1	21.0	23.8	20.9	15.2	9.2	4.2	1.5		
		MCWB	-0.1	-0.2	0.4	4.7	10.7	14.5	17.4	15.9	12.9	8.4	3.8	1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.5	4.3	11.6	16.6	19.9	21.7	20.5	16.3	11.9	7.9	5.4		
		MCDB	2.7	3.3	7.5	15.7	23.3	24.6	27.0	25.6	18.8	12.7	8.4	5.7		
	2%	WB	1.3	1.3	2.7	9.0	14.7	18.5	20.4	18.8	15.3	10.7	6.2	3.5		
		MCDB	1.5	1.7	4.6	12.6	20.0	23.0	25.3	22.7	17.4	11.3	6.6	3.7		
	5%	WB	0.5	0.5	1.5	6.7	13.0	16.9	19.2	17.7	14.2	9.7	5.1	2.1		
		MCDB	0.7	0.8	2.7	10.1	18.3	21.5	24.1	21.6	16.0	10.3	5.4	2.4		
	10%	WB	-0.1	-0.2	0.7	5.0	11.4	15.3	18.2	16.6	13.2	8.5	3.8	1.1		
		MCDB	0.0	0.1	1.6	7.9	16.1	19.8	22.5	20.0	14.8	9.1	4.2	1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.2	7.7	8.3	9.7	8.9	8.8	8.2	6.8	4.6	3.9	4.9		
		MCDBR	4.0	4.2	8.5	11.8	13.7	11.7	11.5	11.0	8.7	5.1	4.0	3.8		
	5% WB	MCWBR	3.9	4.0	6.2	7.1	6.7	5.3	4.8	5.2	5.4	4.3	3.9	3.7		
		MCDWR	4.0	4.3	6.9	11.1	11.6	10.1	10.2	9.6	7.6	4.8	3.8	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.285	0.312	0.351	0.350	0.345	0.370	0.353	0.332	0.312	0.248	0.211		
		taud	2.007	2.112	2.184	2.320	2.453	2.495	2.437	2.468	2.516	2.402	2.098	1.971		
	Ebn at Noon		407	645	783	833	869	880	842	827	778	636	385	231		
		Edh at Noon	34	68	93	101	97	95	98	87	69	53	31	18		
(44) All-Sky Solar Radiation	RadAvg		0.20	0.77	2.28	3.67	4.82	5.31	5.05	3.85	2.13	0.82	0.21	0.09		
	RadStd		0.03	0.18	0.30	0.44	0.55	0.30	0.48	0.45	0.25	0.13	0.04	0.01		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (28 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-191	N/A	N/A	N/A

Nomenclature: See separate page